

Robot / rover

DO

OUTPUT

MOVEMENT

What does it do?

A two-wheel rover uses a separate geared motor for each wheel. A caster or smooth skid supports the other end.

Drive both wheels forwards to move; change their speeds or directions to turn. A motor driver handles their power.

The Robot class joins the two motors into one object. This example uses an Adafruit TB6612 breakout and a separate motor battery pack.

Uses

Robot vacuum cleaner



DETECT
Sense a
nearby
obstacle



DECIDE
Is the path
clear?



DO
Drive on or
turn away

Sock delivery turtle



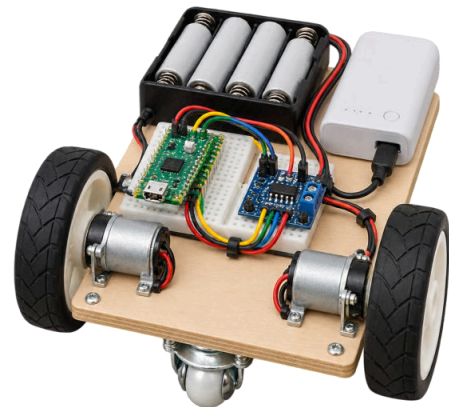
DETECT
Read delivery
button



DECIDE
Is a delivery
requested?



DO
Drive the sock
parcel



Drive wheels

One
geared
motor per
side

Motor driver

Controls
both
motors

Caster or skid

Supports
the
chassis

Power

Separate
motor and
USB
supplies

Code

● Raspberry Pi

```
from gpiozero import Motor, Robot
from time import sleep

robot = Robot(left=Motor(13, 19),
              right=Motor(6, 5))

try:
    robot.forward(0.5)
    sleep(1)
finally:
    robot.stop()
```

● Raspberry Pi Pico

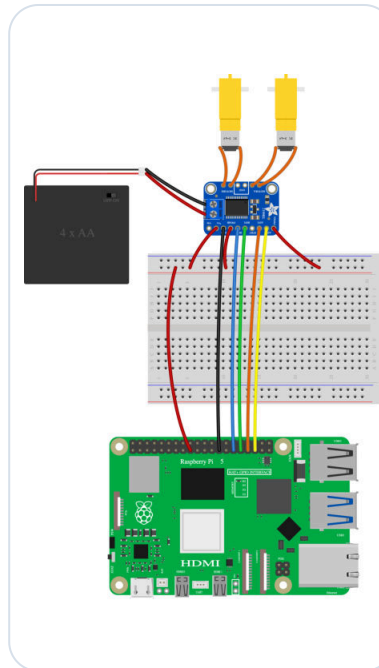
```
from picozero import Robot

robot = Robot(left=(20, 19),
              right=(21, 22))
robot.forward(0.5, t=1, wait=True)
```

Test with wheels lifted. Reverse a motor by swapping its leads with power off. Try left(), right(), backward() and stop().

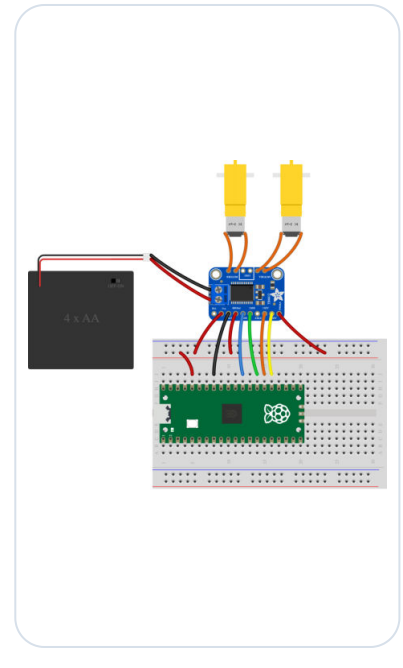
Wiring

RASPBERRY PI 5



GPIO13/19 → AIN1/2 (left); **GPIO6/5** → BIN1/2 (right). **3V3** → VCC, PWMA/B; GND shared.

PICO



GP20/19 → AIN1/2; **GP21/22** → BIN1/2. **3V3** → VCC, PWMA/B. **4×AA alkaline (6 V)** → VM/GND; USB powers board. Use 6 V motors with stall current **≤1 A each**.